

Cyclistics User Behaviour Analysis

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Version: 1

Date: 30/4/2025

Introduction

Every day, thousands of Cyclistic bikes spin across Chicago's streets. Tourists exploring lakefront trails, commuters racing against the clock, and weekend wanderers chasing sunsets.

In user behaviour analysis, the metrics are no longer just mere numbers. Behind a number is a story, a real customer. This case study is going to tell that story for millions of cyclists across Chicago.

Through relentless data cleaning, SQL queries, and precise visual storytelling in R, we dive deep into the heartbeat of Cyclistic's riders, peeling back the layers of when, where, why, and how they ride.

1. Executive Summary

Cyclistic, a bike-share company based in Chicago, seeks to convert more casual riders into annual members. Their leadership has noted that annual members tend to be more profitable than casual members. Casual riders need to be converted to annual members through effective marketing campaigns.

This case study dissects rider behaviour patterns through a full year of trip data (2024), using SQL and R to clean, explore, and visualise insights. Our journey unravelled striking contrasts between casual and annual riders, in trip duration, timing, route choices, and usage cycles, painting a vivid picture of who rides, when, and why.

At the heart of the findings lies a simple story:

"Casual riders ride for pleasure, members ride for purpose."

From there, we developed strategic recommendations designed to target casual riders through targeted marketing campaigns.

2. Objectives

Cyclistic's Leadership wants to create targeted Marketing campaigns to convert Casual riders to Annual Members. For that, we need to answer the following questions:

- **How do casual and member riders differ in their customer behaviour?**
 - Are there specific times, patterns or routes that can help us distinguish them?

3. Data Sources and Tools

Annual data for 2024 was provided by the company, covering a full cycle of business operations. The data contained 6 million rows across every month for 2024.

The data was cleaned and validated through SQL in DBeaver, a database management software. The data was transferred to R, where it was manipulated once again for visualisations. Finally, a Tableau dashboard was created to create live tracking for the important metrics.

4. Data Cleaning and Preparation

Data was meticulously cleaned using SQL:

- **Null Value Checks:** Critical columns like `ride_id`, `started_at`, `ended_at`, and `member_casual` were verified to have no missing values.
- **Logical Consistency:** Trips with negative or zero duration were detected and removed.
- **Duplicate Checks:** `ride_ids` were confirmed to be unique.
- **Blank Stations:** Trips with missing station names were allowed, but noted.
- **Bias Check:** Member rides accounted for ~83% of trips; hence, analyses were separated between casual and member riders

New fields were created:

- **trip_duration** (in minutes)
 - Calculated from the `started_at` and `ended_at` attributes for individual rides.
- **ride_month** (labelled month names)
- **day_of_week** (human-readable weekday)

The final dataset, **all_trips_2024**, combined all months into a single clean table, preserving analytical flexibility across months, allowing us to perform a chronological analysis.

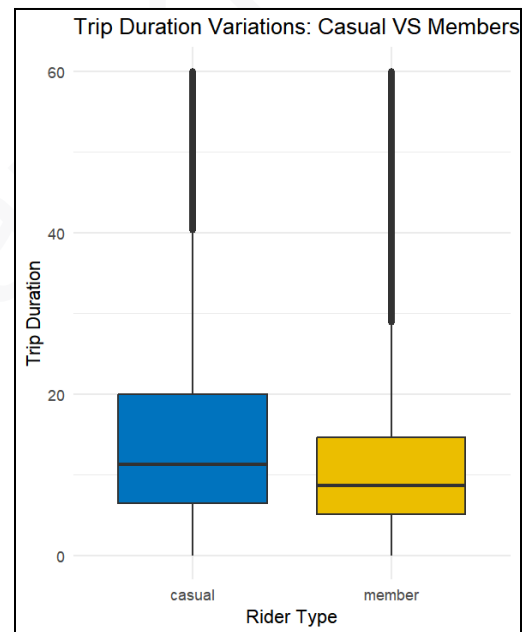
5. Exploratory Analysis and Insights

Who rides for longer?

This is the first clue for us to solve the puzzle. Trip duration was measured overall for Members and Casual riders, which was then plotted on a box plot.

In the graph:

1. The Y-axis represents the trip duration in minutes, ranging from 0 to 60. (Values above 60 exist but for the sake of clarity have been removed from the boxplot, removing 150K observations)
2. The X-axis represents the type of rider.



Insights

The Box plot gives us two compelling insights.

- 1) Casual riders take longer and more varied trips.
- 2) Member riders have shorter, more consistent trip times.

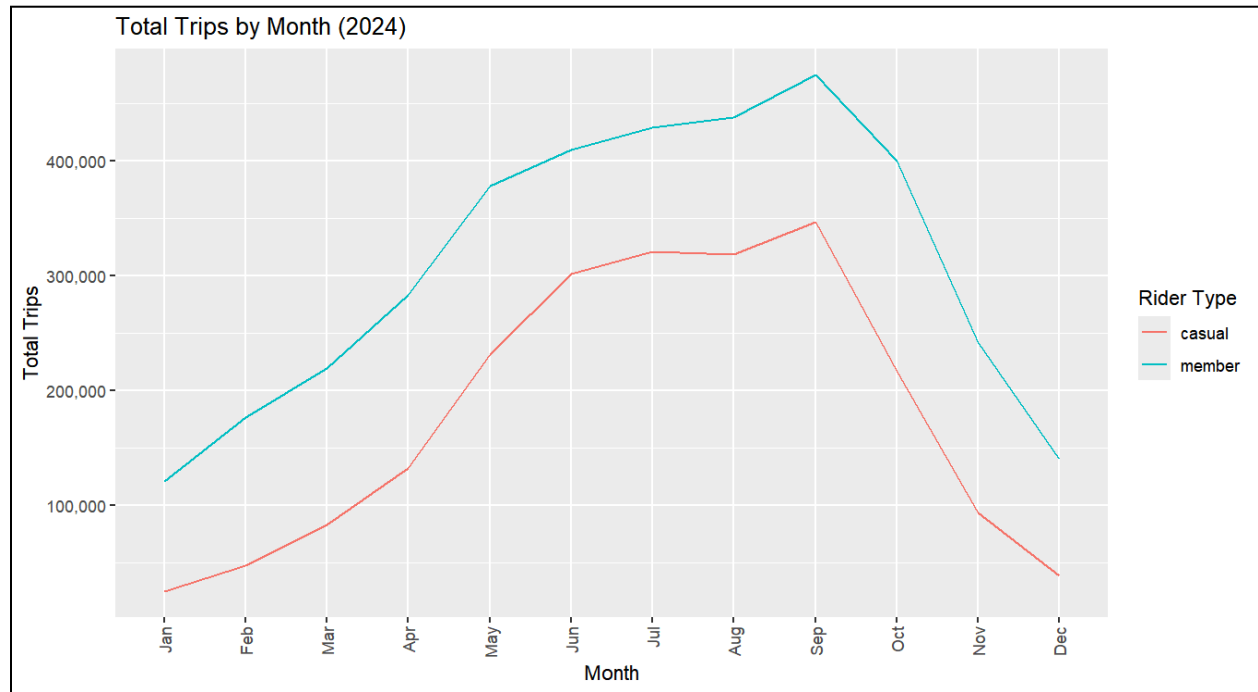
Interesting. It seems that members utilise the service for less time per ride.

One reason for this disparity could be that members tend to use the service for a functional commute, whereas casual riders tend to use it for leisure. Casual riders want to make the most use of the service, while members tend to use it only for transport.

Let's explore further.

Monthly Ride Patterns: When do Most Rides Take Place?

We are interested in identifying whether there is seasonality in the overall ride bookings. Do the casual members and annual members share this seasonality, or, they carry their distinct seasonal heartbeats?



In the graph:

- The X-axis represents the month of the year.
- The Y-axis represents the total trips for each month.
- Each line graph represents the aggregated total for each rider type.

Insights:

- Both casual and member rides **peak in September**.
- For casuals, the peak likely coincides with **pleasant fall weather**.
- For members, it possibly relates to **university sessions restarting**.

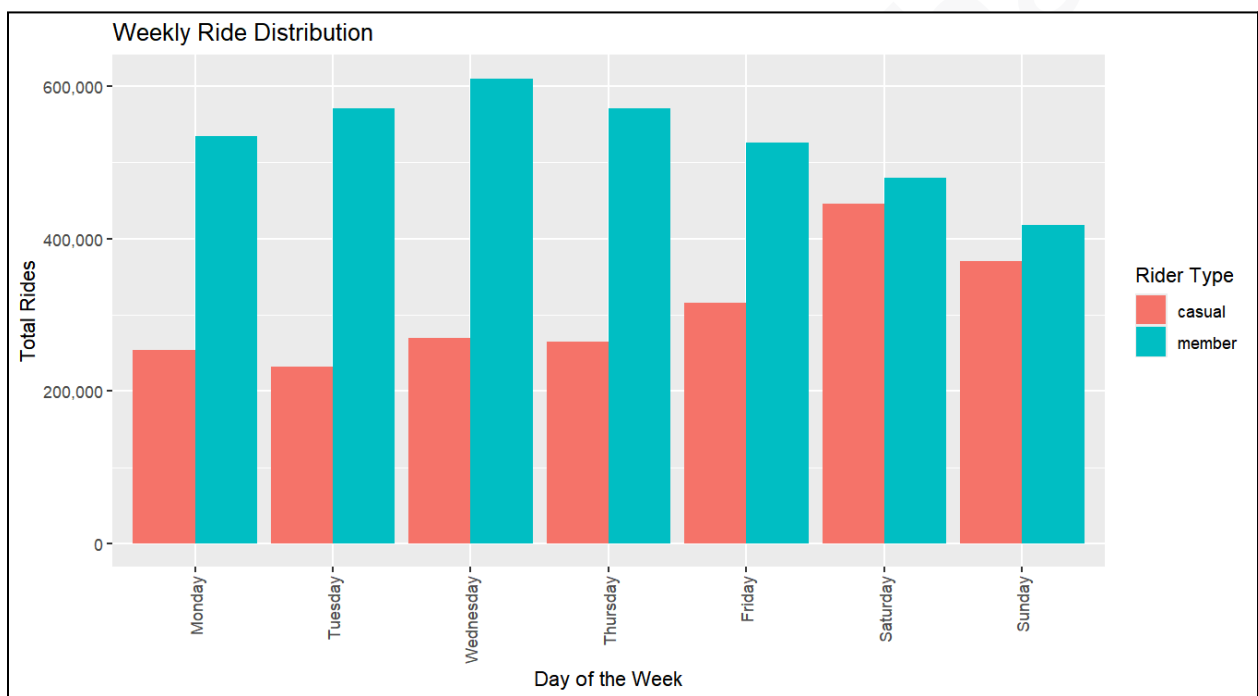
It is quite evident that Casual riders and annual members have the same seasonal curve, both peaking in September. Perhaps, the data can be vetted against Weather summaries in

Chicago to understand whether weather is important in driving this curve.

However, this does not tell us much about the behaviour of the users. Let's look at week-wise ride behaviour for both riders.

Weekly Ride Behaviour: Which day sees the most rides?

An understanding of this is crucial. It would tell us how both users differ on different days of the week. That data can be an essential clue moving us closer to the answer.



In this graph:

- a) The x-axis corresponds to the day of the week
- b) The y-axis is the total rides for each day
- c) Different colours represent the different user types

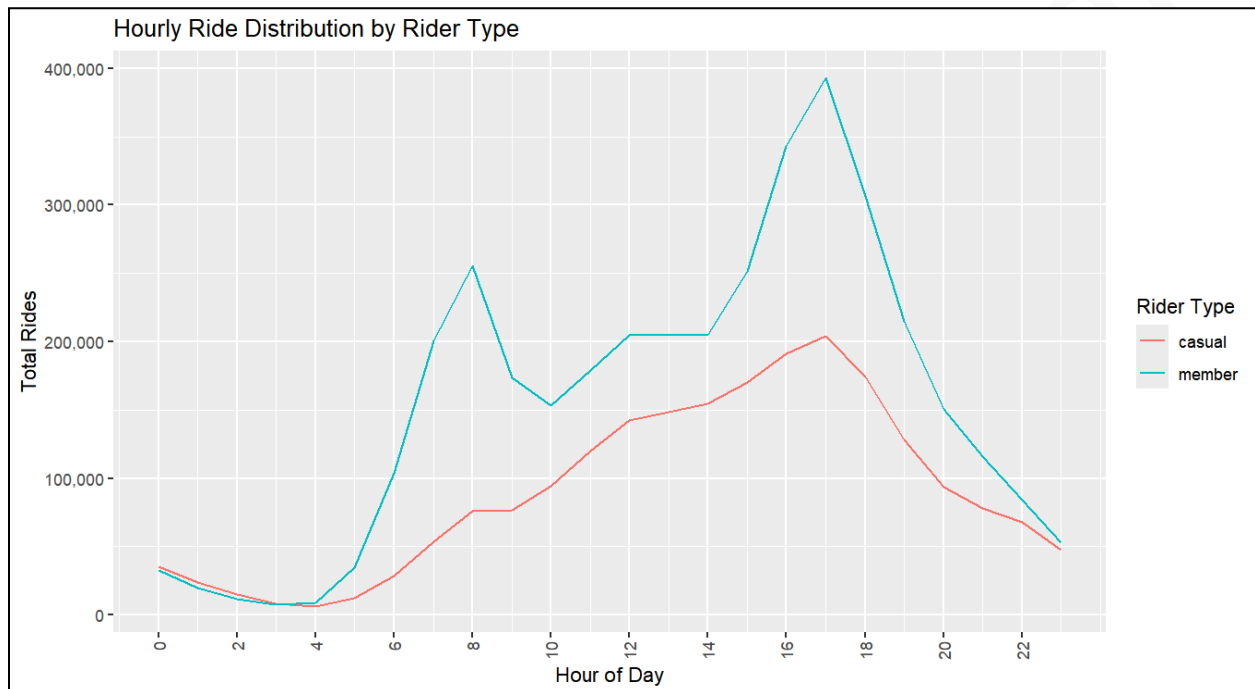
Insights from the graph

- 1. **Members** peak midweek, especially on **Wednesdays**.
- 2. **Casual riders** surge on **weekends**

Both follow an opposite flow. When casual ride declines on the weekdays, member rides tend to peak. Is it a mere coincidence that these days are aligned with normal office days? We need more evidence for a solid argument.

Let's look at the trip frequency at different times of the day.

Hourly Ride Behaviour



In the graph:

- The x-axis corresponds to particular hours of the day.
- The y-axis is the total rides occurring on particular hours.
- Different colored lines correspond to different rider types.

Insights from the graph:

- Member rides spike** twice sharply at **8 AM** and **5 PM**.
- Casual rides rise** gradually, peaking late afternoon, well outside work rush hours.

This pattern is exactly aligned with the initial hypothesis. Members tend to use the service

for commuting, i.e going to work or school and returning. While casual members tend to use it for leisure, as evidenced by the peak in the more pleasant afternoon hours.

By now, it is becoming evidently clear that there exist two distinct user behaviours for the different ride types. However, there is one last clue that can give us even more definite evidence of the user behaviours.

Top Route Analysis: Where do the users even go?

Below are two tables, one for casual riders and one for members. The table highlights the top 5 starting and ending stations for both groups on the basis of total rides between those stations.

Members		Casuals	
Start Station Name	End Station Name	Start Station Name	End Station Name
State St & 33rd St	Calumet Ave & 33rd St	Streeter Dr & Grand Ave	Streeter Dr & Grand Ave
Calumet Ave & 33rd St	State St & 33rd St	DuSable Lake Shore Dr & Monroe St	DuSable Lake Shore Dr & Monroe St
Ellis Ave & 60th St	University Ave & 57th St	DuSable Lake Shore Dr & Monroe St	Streeter Dr & Grand Ave
Ellis Ave & 60th St	Ellis Ave & 55th St	Michigan Ave & Oak St	Michigan Ave & Oak St
University Ave & 57th St	Ellis Ave & 60th St	Millennium Park	Millennium Park

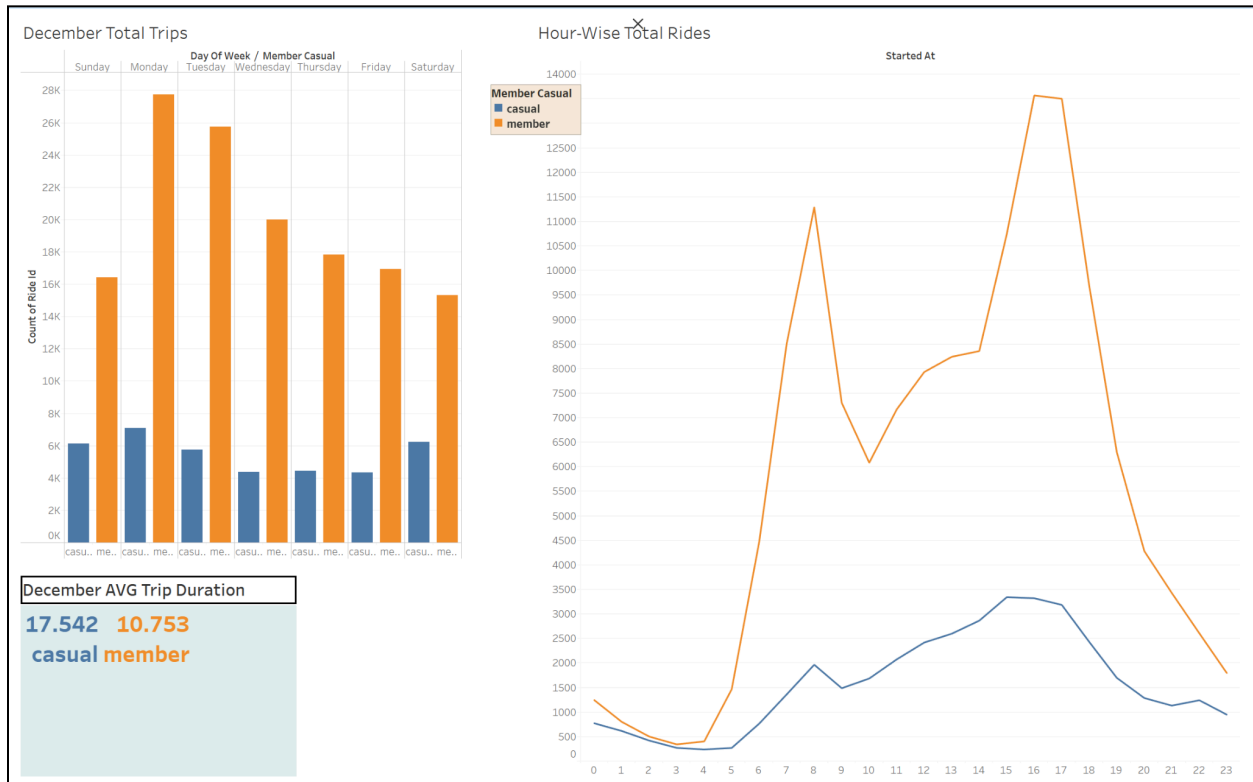
Before we do any geographical analysis, just by looking at the station names, important insights emerge.

For the members, the stations in positions 1 and 2 are the same. That means it is a regular route where people go to either work or school and return home on the same route. Similarly, routes 3 and 5 are also the same route.

For casual riders, some starting and ending station names are the same. This means, casual

riders rode around and returned to the same station. Hence, they used the service for leisure, just enjoying the day out instead of going anywhere.

Tableau Dashboard



The Tableau dashboard is meant to provide live tracking of important statistics giving us a diagnostic look of where in the season we are. In this dashboard, we are looking at three metrics that is tracking avg trip duration, total trip by day of week and total trips by Hour segmented for both user types.

In this current state, the dashboard is showing the monthly details for December, since there is no live dataset to be connected to it.

Summary

We have ample evidence to map out the differences in behaviours for both casual and member riders. Casual riders use the service for leisure, while members use the service for

commuting. Here is a summary of the arguments:

- Casual riders have a higher trip duration on average than non-casual riders.
- Casual riders prefer to ride on weekends rather than weekdays.
- Casual rides peak in the late afternoon period.
- Member rides peak during rush hours.
- Casual riders circled back to their starting station while members preferred one-way trips.

Business Recommendations

The goal of this analysis was to find out how we can convert the casual riders to the more profitable member class. The short answer is, you cannot. There is a significant difference in their user behaviours to be targeted for conversion. Rather, to target casual riders, new services would have to be launched specifically targeting casual riders. Here are my recommendations:

- 1. Create Tourist Bundles:** Rather than giving one-day passes, it would be more profitable to cover more days in a single bundle, allowing tourists or leisurely users to travel without worry.
- 2. Proper Channel Targeting:** Advertise casual bundles on tourist-heavy websites like hotel bookings and trip bookings.
- 3. Targeted SEO Ranking:** Rank for top tourist stations that are used frequently by casual riders.
- 4. Tourist Stations Discounts:** Provide discounts on specific starting stations based on casual usage for a limited period.
- 5. Push Notification Targeting:** Continuous targeting of frequent casual riders, highlighting cost savings for Casual Rider Bundles or even a discount on annual memberships.
- 6. Conduct Customer interviews** with casual riders and understand why they don't prefer annual memberships.

Conclusion and Further Analysis

At the heart of every company are the customers. It is for them that the company is solving a problem, making their life one bit easier. Through this analysis, we peered into the lives of Cyclistics customers, figuring out how they ride, when they ride, and where they go. All this information is used to make their life one bit better than it already is.

However, this analysis can be taken further in many different ways. Such as:

1. What will happen if you introduce hourly charges for casual riders?
2. Do casual users behave like annual members by frequently using the service?
3. How can we expect the overall base of customers to increase by new casual and annual members?

Depending on the business focus, any of these objectives can be tapped into.